



Hand me a wrench, Robot

ARMAR-6 is a new maintenance assistant robot which can understand simple commands and get you the right tool. <http://digit.in/RobtHelp>



"Will come to a bad end"

Berkshire Hathaway CEO, Warren Buffet believes that cryptocurrencies will come to a bad end. <http://digit.in/BadCrypto>

the loop, controlling the robot remotely, and provide a full body teleoperation platform. While performing the actions, such as using an axe to break through a wall, a normal robot usually falls after performing the action even once. With HERMES, a metal exoskeleton attached to the waist translates information from sensors on the robot specifically designed to measure the changes in the weight distribution of the robot. The human operator can feel the shift in weight, and adjust their stance accordingly, letting the robot perform momentum driven tasks. Through the use of the bilateral feedback, the robot moves just like the human in real time.

An approach by researchers at the University of Purdue was to use a Kinect camera from Microsoft to record a human performing actions, and then map the actions to a Hubo II – an android manufactured by Rainbow Robotics. Through this approach, the robot was taught the graceful motions of Tai Chi, as well as to dance to Gangnam Style. While the arm movements were accurate renderings, the robot was not able to move its legs in the way that Psy does.

Most androids lean forward at the waist and bend their knees to keep their balance while walking. The gait is not human like, as the robots tend to place the entire foot on the ground at once, before moving the other leg. Researchers from Georgia Tech have given a robot a more natural gait by allowing it to place its heels first on the ground, then roll forward, and push off with the tip of its toes. This gait resembles the human manner of walking more, allowing the team to significantly reduce the amount of energy it takes for the robot to walk. The robot was even given a pair of shoes meant for humans, and the next step for the team is to make the robot climb stairs, and run across uneven fields.

The Atlas platform by Boston Dynamics is one of the most advanced humanoid robots in existence. The robot balances by taking the input from multiple sensors in its body and legs. Atlas was used by no less than six teams appearing in the DARPA robotics challenge finals. Even though Atlas

has bent knees and a flat footed gait, it can carry out a lot of tasks. The robot can navigate uneven terrain, stand on one leg, jump between platforms, perform backflips and even open doors.

GETTING A GRIP

Robotic hands have traditionally been problematic for a number of reasons. They are not nearly as dextrous and flexible as human hands. The human hand itself is anatomically one of the most complicated areas in the human body. While robotic hands can replicate this complexity by using a number of sensors and motors, these kinds of hands would be complicated to pro-



A soft robotic glove that restores lost motor abilities in the hand. Credit: Wyss Institute at Harvard University.

gram, and so expensive that they would be impractical. Researchers from Yale and Harvard approached this problem by taking inspiration from another creature – the humble cockroach. The flexible and springy legs of the cockroach allows them to navigate tricky surfaces, and using the same approach for robotic graspers also increased their efficiency at holding objects of various shapes and materials. The advantage of this approach is there are fewer motors and sensors, reducing the weight of the robotic hand for use in prosthetics.

One promising area of research that can have solutions is the field of

soft robotics. The cold, hard hands of robots are not very great at say, delicately handling pets. Researchers from Harvard have developed a soft robotics glove that can help patients with loss of motor abilities in their hands. The robot is just like a glove, and fits easily over a hand, improving the gripping capability of the patient. The glove is made up of soft multi-segment actuators, that supports the entire range of motions possible with the human hand.

Engineers at Riken, a research institute in Japan, in collaboration with Sumitomo Riko Company have developed the Robear, a medical robot with the appearance of a teddy bear. The feedback from the actuators in realtime, while performing the tasks, allows the robot to handle patients gently, even when performing power intensive tasks, such as lifting them. The legs have extensions that come out when carrying a patient, for stability, but retract when the robot is navigating tight spaces.

A challenge that robotics in prosthetics is to allow the human brain to manipulate the appendages. Researchers from Duke University have successfully demonstrated monkeys manipulating robotic arms using signals captured from their brains. Researchers from MIT and Boston University have developed a system that allows a human to correct mistakes made by a robot through an ECG interface. An AI process the brain activity recorded by the ECG monitor and passes on the instructions to the robot. Such research in the future could allow for the development of neuroprosthetics, and independent robots that can be controlled with the brain.

Researchers from Georgia Tech have experimented with a smart arm for a drummer. Attached to the shoulder, it allows the drummer to actually have three arms. The robot responds to human gestures, as well as the music that it hears. The arm keeps speed with the tempo of the drummer, and dynamically moves to parts of the drum kit that the drummer is not playing. Robotic arms can do much more than just restore lost capabilities, and can augment existing capabilities. 